

Rice Price Dataset

This dataset focuses on understanding Rice price patterns and agricultural market trends using variables such as Rice type, Region, price per kg, Demand, Supply, Rainfall, Temperature, Import, Export and Date.

Data Preprocessing

- Remove Duplicates (20 duplicate values are found)
- In column Rice ID values were not arranged in sequential order so, I sorted into smallest to largest.

Missing Values Handling

Missing values are identified using COUNTBLANK function

- Missing values in the **Rice ID, Region, Rice Type, Import & Export** columns were identified using the Go To Special → Blanks option and filled with 'Unknown'.
- **Price per kg:** Missing values were handled using Median Imputation.
- **Demand Tons:** Missing values were handled using Median Imputation.
- **Supply Tons:** Missing values were handled using Mean Imputation.
- **Rainfall mm:** Missing values were handled using Mean Imputation
- **Temperature C:** Missing values were handled using Mean Imputation.
- The Data Type of the Numerical columns were converted to Number/Decimal Number for proper Analysis
- **Date:** 15 missing dates were identified using the COUNTBLANK function and manually filled based on the data sequence, Converted Data type into Date format.

Inconsistency Handling

- In Region column:
NORTH → North
south → South

- In Rice type column:
basmati→ Basmati
jasmine→ Jamine
- In Import & Export column:
EXPORT→ Export

import→ Import

I used Find & Replace method to handle inconsistency.

- In Region column I found some inconsistency like unwanted space in the string so I use CLEAN & TRIM function to handle.

Pivot Tables

- Region wise Demand & Supply.
- Import v/s Export Distribution.
- Total Demand by Rice type.
- Average Rainfall by Region
- Average Rice Price by year and Rice type.
- Average Temperature by Rice type.

Power BI Data Visualization

At first I import cleaned Excel file into power BI. During the import, the Price ID column contained some blank values that were replaced with “unknown” in Excel. But price ID is a numeric column, Power BI identified the text value “Unknown” as an error. Therefore I used power query editor to identify the errors and replaced those errors with “unknown” to handle the missing data appropriately.

Created Calculated Columns

- Price Category→ Categorized rice price into Low, Medium & High based on the price range in dataset.

- Supply Demand Status→ Supply & Demand values were compared to classify each record as Supply Surplus, Supply Deficit & Balanced.
- Weather Condition→ Rainfall & Temperature were combined to classify the weather into different conditions such as Cool & Dry , Hot & Dry, Cool & Rainy, and Hot & Rainy, based on the value ranges in the dataset.

These 3 calculated columns are created using DAX IF function.

Rice Price Analysis Dashboard

1. Rice Price Overview

- Insert 3 Slicers→ Year, Rice type, Region
- Use card visual to create KPI's → Average rice price, max rice price, min rice price & total records.
- Created Rice price trend overtime using→ Line chart.
- Created Average price by rice type using→ Clustered column chart.
- Created Price category distribution using→ Donut chart.
- Created Average rice price by region using→ Stacked bar chart.

2. Market & Demand Analysis

- Insert 3 Slicers→ Year, Rice type, Region
- Use card visual to create KPI's → Total Demand, Total Supply, Demand Supply Gap
- Created Supply – Demand Status by Region→ Stacked Bar Chart
- Created Total Demand by Region using→ Clustered Column Chart
- Highlights Highest Demanded Rice using → Card Visual
- To find relation between Average rice price & Total Demand in Rice type using→ Scatter chart

3. Weather & Price Analysis

- Inset 3 Slicers→ Year, Rice type, Region.
- Use card visual to create KPI's → Average Rainfall, Average Temperature & Total Rice Price.
- Created Average Temperature by Region using→ Stacked Column chart
- Created Average Rainfall by Month using→ Line chart.
- Created Price Variation Across Weather & Region using→ Clustered Column Chart.

4. Insights & Recommendations

- Use card visual to create KPI's→ Top rice, Top Demand, Low Demand.
- Created a Decomposition Tree to analyze Average Rice price based on Rice type, Region, Demand, Supply, Rainfall & Supply.

Interactive Features

- Added a Reset Button to each dashboard pages.
- Created Page Navigation by using Buttons.
- Created a Drill through page based on Rice type to provide detailed analysis of Price, Demand, and Supply by month.

Note: Records with “Unknown” values were filtered out from analysis to ensure clearer and more meaningful insights.